

Business Entity Resolution (ER)

1. Objective

The Amazon ML Challenge 2026 focuses on **Business Entity Resolution (ER)**, where business records from three independent sources contain partial and noisy information about the same real-world entities. Since the records do not share a common identifier, the task is to determine which records from different sources correspond to the same business using attributes such as **business names and addresses**.

Source 1 (S1) serves as the deduplicated reference source. For each S1 entity, the system must identify and output all corresponding records from **Source 2 (S2)** and **Source 3 (S3)**. An S1 entity may have **zero, one, or multiple matches** across S2 and S3. Entities for which no corresponding record is found are treated as **singletons** and must still be included in the output with an empty match list.

2. Data Description

Each source file contains four columns: **entity_id**, **business_name**, **business_address**, and **country**. The entity_id prefix (S1-, S2-, or S3-) identifies the corresponding source. All datasets are provided in **tab-separated (.tsv) format**, as business names, addresses, and entity ID lists may contain commas.

The training ground-truth file, **train_ground_truth.tsv**, maps each source1_entity_id to a comma-separated list of its corresponding matched_entity_ids.

Split	S1 Records	S2 Records	S3 Records
Train (labelled)	22,06,821	50,34,616	52,85,603
Test (unlabelled)	17,32,544	48,87,273	50,82,316

The training data covers the US and India. The test set adds a third country, France, that never appears in training (test S1 by country: India 809,986, US 663,106, France 259,452). The rules require country to be treated as an open set of labels: no hard-coding, filtering or one-hot encoding of {US, India}, and every test entity (France included) must appear in the submission.

3. Methodology

3.1 Overview

The problem is formulated as **pairwise classification over a pruned candidate set**, followed by a **set-level assignment step** designed to optimise the per-entity **F0.5 score**. The overall architecture of the proposed entity-resolution pipeline is illustrated in **Fig. 1**. The complete pipeline consists of six stages:

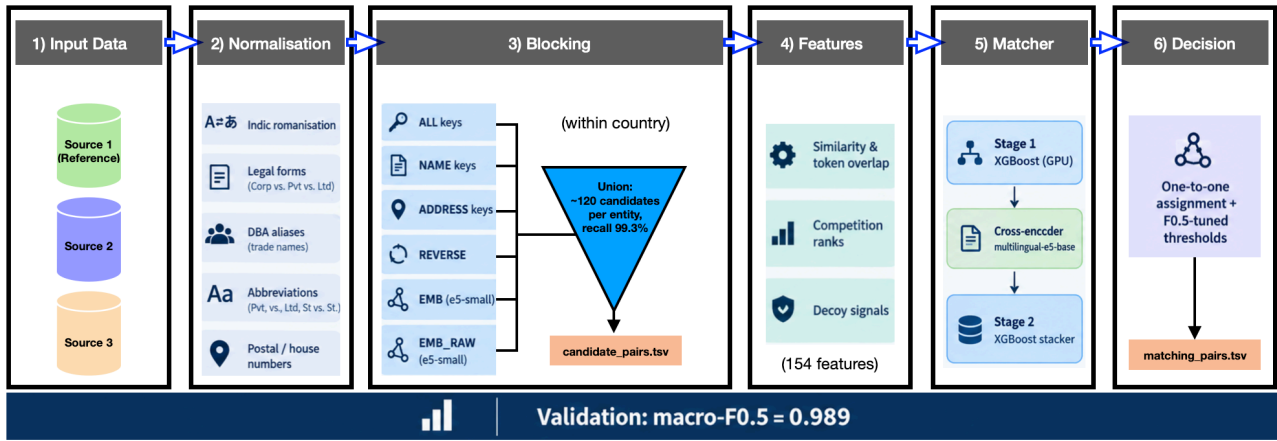


Figure 1. Overall architecture of the proposed business entity-resolution pipeline.

- **Input Data:** Three independent sources—**S1, S2, and S3**—provide business records containing business names, addresses, and country information. **S1 serves as the reference source**, while S2 and S3 contain records to be resolved against it.
- **Normalisation:** Business records are normalised through Indic romanisation, legal-form standardisation, DBA alias extraction, abbreviation handling, and postal/house-number processing to produce consistent representations for matching.
- **Blocking:** Candidate pairs are generated using six retrieval passes: ALL keys, NAME keys, ADDRESS keys, REVERSE retrieval, EMB, and EMB_RAW. Their union produces approximately 120 candidates per entity with 99.3% pair recall.
- **Features:** Each candidate pair is represented using 154 features, including similarity and token-overlap measures, competition ranks, and label-free decoy signals.
- **Matcher:** **Stage 1** uses a GPU-based XGBoost classifier, followed by a **multilingual cross-encoder** for promising candidates. A **Stage 2** XGBoost stacker combines the Stage-1 probabilities, cross-encoder scores, and ranking information.
- **Decision:** Final matches are produced through **one-to-one assignment** and **F0.5-tuned** thresholds, generating the final matching_results.tsv

Three principles guide the overall design:

- **Country-agnostic processing:** No country-specific rules are used. The same pipeline operates across all country labels, with statistics recomputed independently for each split. Monotone constraints also prevent the trees from learning US- or India-specific patterns, allowing the unseen French data to be processed automatically.
- **Explicit competition modelling:** Many errors arise between highly similar businesses. The model therefore captures how a candidate ranks relative to other candidates for the same S1 entity, as well as how an S1 entity ranks among other S1 entities competing for the same candidate.
- **Leakage-safe scoring:** Every score used for decision-making is generated from held-out data. **Grouped out-of-fold predictions, a cross-fitted cross-encoder, and a 10% entity-level holdout** are used to prevent information leakage and maintain an unbiased validation estimate.

3.2 Pre-processing and Normalisation

The pre-processing uses a compact set of hand-crafted canonicalisation maps, together with a single dictionary learned from the training ground truth. The same normalisation functions are applied to both records in every candidate pair, making **consistency of transformation** more important than linguistic perfection.

a) Shared Cleaning

Text written in Brahmic scripts—including **Devanagari, Bengali, Gurmukhi, Gujarati, Odia, Tamil, Telugu, Kannada, and Malayalam**—is romanised using a shared Unicode offset table. For example, "सूर्या प्राइवेट लिमिटेड" is converted to "*surya praivet limited*".

Accented characters are normalised using **NFKD decomposition**, text is lower-cased, apostrophes are removed, and & and + are converted to and.

b) Business Names

Business name normalisation consists of the following steps:

- Website domains are reduced to their root stem, while common OCR-style digit substitutions are corrected.
- DBA, AKA, *trading as*, and *formerly* markers are used to separate the primary business name from aliases. Each alias receives independent blocking keys and contributes an `alias_best` feature.
- Dotted abbreviations are merged, while canonicalisation maps shorten common legal forms and descriptors such as *Corporation* → *corp*, *Private* → *pvt*, etc.
- A **token transliteration map** is learned from true training pairs where Indic- and Latin-script names have matching token counts. Token mappings are retained only when they occur at least three times and achieve a 50% majority.
- Legal forms, and generic edge descriptors are removed from both ends of the name.
- Instead of discarding information, the processed name is represented through four fields: **core**, **legal**, **alias**, and **extra**.

c) Addresses

Address normalisation includes:

- Standardising ZIP+4 and split PIN codes.
- Canonicalising landmark phrases, removing labels such as *pin code* and *zip*, and standardising selected city and state names.
- Mapping English and French street types to canonical abbreviations, while removing pure numbering prefixes such as *No.*, *#*, *Plot*, *Door*, and *Flat*.
- Extracting postal codes, normalising numeric tokens, and recording whether an address is landmark-based.

Additional derived representations include a **phonetic skeleton** designed to remain relatively stable across transliteration variants, sets of numbers appearing in the name and

address, the first house number, and an acronym. All normalised profiles are cached separately for each data split.

3.3 Candidate Generation / Blocking Strategy

Blocking determines the **recall ceiling** of the entity-resolution pipeline. To maximise candidate coverage while keeping computation tractable, the system combines multiple retrieval passes and takes their **union**. Blocking is performed independently within each country label. The groups are constructed dynamically from the labels present in each split, allowing an unseen country such as **France** to form its own group in the test set without any country-specific logic. Records with an empty country label are treated as **wildcards**.

a) Weighted Blocking Keys

Each record is represented by a collection of **typed blocking keys**. Within each country group, each key receives an IDF-style weight:

$$w(k) = \log \left(1 + \frac{N}{df_{S1}(k) + df_{S2/S3}(k) + 1} \right) \times \alpha_{\text{type}} \dots\dots\text{eq(1)}$$

where N is the number of records in the relevant group, df denotes the key frequency, and α_{type} is a multiplier specific to the key type.

Keys that occur on only one side of a comparison are discarded.

Keys whose implied candidate count, $|S1_k| \times |S2/S3_k| \dots\dots\dots\text{eq(2)}$

exceeds **2 million pairs** are also removed to prevent overly common keys from dominating retrieval. The remaining records are represented as **L2-normalised sparse vectors**, making the sparse dot product equivalent to a weighted cosine similarity. DBA, AKA, and *formerly* aliases generate their own name keys, allowing records listed under a trade name to remain retrievable even when the primary business names differ.

b) Retrieval Passes

Six complementary retrieval passes are used:

Pass	What it searches	Depth	Why it is needed
ALL	Every key type	top 20 per S1, per source	Main pass.
NAME	Name keys only	top 10	Finds records with a missing or very different address.
ADDRESS	Address keys only	top 15	Finds rebranded names or names written in another script.
REVERSE	ALL keys, from the S2/S3 side	top 3 S1 per S2/S3 record	Rescues records whose S1 is crowded out by look-alikes.
EMB	multilingual-e5-small on normalised name + address	top 20	Semantic / fuzzy similarity beyond shared keys.
EMB_RAW	multilingual-e5-small on the original text	top 20	Reads native scripts directly instead of their transliteration.

The sparse retrieval passes use multi-threaded top-k sparse matrix multiplication through **sparse_dot_topn** (Apache-2.0), with a SciPy fallback. Computation is divided into chunks according to workload and distributed across CPU processes. Each retrieval pass is executed independently for **S2 and S3**, preventing one source from crowding out the other.

For the embedding passes, each record is encoded once in **FP16** on the GPU, followed by cosine top-k retrieval using PyTorch. The outputs of all six passes are merged into a single candidate set. For every candidate pair, the pipeline retains a **bitmask indicating which passes retrieved the pair**, together with cosine similarities from the ALL, NAME, ADDRESS, and embedding spaces. These retrieval signals are subsequently used as model features.

The resulting union is written to **candidate_pairs.tsv** and is exactly the set scored by the Stage 1 and Stage 2 matching models. Consequently, every final matched ID is guaranteed to have appeared in the candidate-generation stage.

c) Blocking Results

The individual retrieval passes are complementary: no single pass achieves a pair recall above 0.98, but their union reaches 0.9933 because the different passes recover different subsets of true matches.

On the training set containing **2,206,821 S1 entities**, the blocking results were:

Metric (train, 2,206,821 S1)	Value
Pair recall of the union	0.9933
Oracle macro-F0.5 ceiling (perfect matcher on these candidates)	0.9980
Recall by pass: ALL / NAME / ADDRESS / REVERSE / EMB / EMB_RAW	0.9765 / 0.6084 / 0.8610 / 0.9562 / 0.9671 / 0.9670
Recall of ALL pass at depth 1 / 3 / 5 / 10 / 20	0.4667 / 0.8809 / 0.9474 / 0.9666 / 0.9766
Average candidates per S1	120.6 (266.2M pairs)
Reduction ratio vs. all S1 x S2/S3 pairs	about 0.999988

On the test set, blocking produced **213,023,010 candidate pairs**, corresponding to approximately **123 candidates per S1 entity**. The candidates were distributed across countries as follows:

- **France:** 33.2M pairs
- **India:** 103.3M pairs
- **US:** 76.5M pairs

This represents a reduction ratio of approximately **0.999988** relative to the full **1.73M × 9.97M** S1 to S2/S3 cross-product, while retaining **99.33% of true matching pairs** in the training evaluation.

3.4 Feature Engineering

3.4.1 Stage-1 Features

It uses **154 features per candidate pair**, designed to capture both direct similarity and the competitive context of each candidate.

a) *Global Context Features*

These features (58) describe how a candidate compares with other candidates in the same retrieval set. They include blocking and embedding cosine scores, token-set similarities, retrieval-pass indicators, candidate ranks, and score gaps. For each S1 entity, the candidate's rank and gap to the best and second-best candidate are calculated. The same statistics are computed from the reverse perspective, comparing S1 entities competing for the same S2/S3 record. Group sizes are also included. This helps distinguish a clear best match from several highly similar alternatives.

b) *Per-Pair Features*

These features (83) directly compare the two business records. They cover name and address similarity, including **RapidFuzz/Jaro-Winkler** measures, token overlap, phonetic-skeleton similarity, and name-in-address checks. Additional features capture **IDF-weighted token overlap, postal-code and house-number agreement, numeric conflicts, aliases, acronyms, legal forms, landmark information, and token statistics**. These features help identify both strong similarities and informative differences between candidate pairs.

c) *Label-Free Decoy Features*

Decoy features are designed to detect look-alike businesses without using ground-truth labels. They capture house-number differences, small number shifts, number-support statistics among competing candidates, and counts of records sharing the same name or name-plus-address combination. The system also learns decoy-word statistics independently within each country and split. Words that frequently appear in near-identical records with shifted house numbers receive higher decoy scores. Because these statistics are learned from the split itself, the method can also discover such patterns for an unseen country like France.

The Stage-2 stacker combines the Stage-1 probability and cross-encoder score with their availability flag, candidate ranks and score gaps, blocking and embedding similarities, group sizes, decoy signals, and source indicators. Together, these features allow the stacker to combine **pair-level similarity, retrieval evidence, and competition context** before making the final matching decision.

3.5 Model Architecture

a) *Stage 1: Gradient-Boosted Pair Classifier*

Stage 1 uses XGBoost 2.1.x on GPU with a binary-logistic objective. The model uses depth-8 trees, learning rate 0.05, subsampling and column sampling of 0.8, and 256 histogram bins. Monotone constraints enforce that higher similarity and smaller gaps cannot reduce match probability, while conflicts, unmatched-token signals, and ranks receive the opposite constraint.

Training uses **600,000 sampled S1 entities**, with positive pairs and selectively sampled hard/easy negatives. Three entity-grouped CV folds are used for leakage-safe evaluation, with early stopping.

b) *Cross-Encoder*

The second component uses int float/multilingual-e5-base with a classification head. It jointly reads the original name and address of the S1 record and candidate, allowing it to capture information that normalisation may remove, including native scripts, contextual words, and number formats.

It is trained using binary cross-entropy with **AdamW**, learning rate 2×10^{-5} , batch size 128, one epoch, and BF16. To control leakage, the model is **cross-fitted across two halves of the training entities**. It scores the top candidates selected by Stage 1, together with candidates exceeding the Stage-1 probability threshold.

c) *Stage 2: Stacking Model*

A second XGBoost model combines the Stage-1 probability, cross-encoder score, competition statistics, blocking signals, and other Stage-2 features. It uses the same general parameters and monotone constraints as Stage-1.

The model is trained using the same entity-grouped folds and row-sampling strategy. Its best CV rounds were **287 / 258 / 268**, with a validation log loss of approximately **0.0019**, compared with **0.0031** for Stage 1. The final model is refitted for **298 rounds**. Importantly, Stage-1 probabilities supplied to Stage 2 are always out-of-fold or generated by a model that has not seen the corresponding S1 entity, preventing information leakage.

4. Training and Validation Strategy

- **10% entity holdout:** 220,682 training S1 entities are held out from all training and tuning. The final validation score is measured exclusively on these entities.
- **Grouped cross-validation:** Data is split by S1 entity so that all candidates belonging to the same entity remain in the same fold. Stage-1 and Stage-2 predictions for training entities are therefore generated out-of-fold.
- **Refit models:** Entities outside the 600k training sample, including the holdout set, are scored using models that have not seen those entities during training.
- **Full competition:** Competition features and one-to-one assignment consider all training pairs, ensuring that validation experiences candidate crowding similar to the test set.

5. Decision Layer

Candidate probabilities are converted into final match lists through two steps:

a) **One-to-one assignment:** S1 is deduplicated so each S2/S3 record can be assigned to at most one S1 entity. A candidate is retained for the S1 entity giving it the highest probability.

b) **Match selection:** Thresholds are tuned on out-of-fold training predictions. The system

evaluates separate S2 and S3 thresholds using a coarse-to-fine search over **0.20–0.95**. An alternative expected-F0.5 rule selects the top-k candidates whose expected F0.5 is maximised under a Poisson-binomial model. Both strategies are evaluated using Stage-1, Stage-2, and blended probabilities.

The final threshold of **0.775** reflects the precision-oriented nature of F0.5. Predictions therefore require high confidence, while an empty match list remains the default for uncertain entities.

6. Handling the Unseen Country: France

The pipeline treats country as an **open-set string label** rather than a fixed categorical variable. Country is used only for blocking groups and a same-country feature; it is not one-hot encoded or hard-coded.

French-specific linguistic patterns are supported through the general normalisation system, including legal forms and address terms such as *sarl*, *sas*, *compagnie*, *societe*, *rue*, and *chemin*. IDF statistics and decoy-word tables are recomputed on each split, allowing French-specific statistics to be derived directly from the test data. The multilingual embedding and cross-encoder models can process French text, while monotone constraints reduce the risk of country-specific memorisation. An optional calibration procedure can additionally adjust predictions for an unseen country. In the final run, France's predicted match rate changed from **0.9457 to 0.9430** after applying a logit bias of **−0.807**.

7. Results

The proposed pipeline achieves a **holdout macro-F0.5 of 0.98908** on 220,682 S1 entities that were never used during training or tuning. This is closely aligned with the out-of-fold score of **0.98915**, indicating consistent performance between training validation and the held-out entities.

Measure	Value
Holdout macro-F0.5 (220,682 never-trained S1)	0.98908 (US 0.98872, India 0.98961)
Out-of-fold macro-F0.5 on all 2,206,821 train S1	0.98914 (US 0.98877, India 0.98969)
Out-of-fold macro-F0.5 on training entities only	0.98915
Oracle ceiling given the candidates	0.9980
Test S1 rows / with matches / empty	1,732,544 / 1,634,463 / 98,081
Official validator	PASS

The remaining gap to the **0.9980 oracle ceiling** mainly comes from the matching and decision stages rather than candidate generation. Stage-2 stacking with the cross-encoder improves the final performance by approximately **0.007** over Stage-1 probabilities alone (**0.98923 vs. 0.98202**).

8. Limitations and Future Work

- **Limited training diversity:** Only one booster seed and three CV folds were used due to computational constraints. Multiple seeds and five-fold CV could provide more robust estimates and potentially improve performance.
- **Unlabelled France:** Since the French test data has no ground-truth labels, its quality can only be assessed indirectly through match-rate consistency and calibration. A small labelled French subset would enable direct evaluation.
- **Transliteration-map leakage:** The 505-token transliteration dictionary is learned from the complete training ground truth, including the holdout entities. Although the expected impact is small, learning the dictionary exclusively from non-holdout entities would make the validation protocol fully label-clean.
- **Remaining improvement opportunities:** Approximately **0.67% of true pairs are missed during blocking**, while the final score remains below the **0.9980 oracle ceiling**. These represent the main opportunities for future improvement. Concrete cases are recorded in `blocking_misses.tsv` and `oof_errors.tsv`.